

Machine Learning for Emergency Department Length of Stay, Admission, and Hospitalization Prediction Using Triage-Only Data

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ABSTRACT

Background: Emergency Department (ED) crowding is a critical public health challenge. Prolonged boarding (≥ 8 hours) affects over 20% of visits. Hospital admission and prolonged hospitalization (> 7 days) consume disproportionate resources. Clinicians currently rely on triage scales like the Emergency Severity Index (ESI), which have limited predictive accuracy. Machine learning (ML) using triage-only data could support these decisions, but evidence on model accuracy remains limited.

Methods: Using the MC-MED dataset (118,385 adult ED visits), we evaluated four ML models (XGBoost, LightGBM, Random Forest, TabNet) on three binary classification tasks using 29 admission-available variables: (1) predicting prolonged ED length of stay (ED-LOS ≥ 8 hours vs. < 8 hours), (2) predicting hospital admission (admit vs. discharge), and (3) predicting prolonged hospital length of stay (hospital LOS > 7 days vs. ≤ 7 days) among admitted patients. All 29 features were available within 90 minutes of arrival. Area under the receiver operating characteristic curve (AUC-ROC) was the primary performance metric. SHAP (SHapley Additive exPlanations) analysis identified the most important features for each prediction task.

Results: For ED-LOS ≥ 8 hours, XGBoost achieved an AUC-ROC of 0.660 (95% CI: 0.652-0.668). For hospital admission, XGBoost achieved an AUC-ROC of 0.817 (95% confidence interval [CI]: 0.810-0.824), an improvement over the ESI baseline of 0.678. For hospital LOS > 7 days among admitted patients, Random Forest achieved an AUC-ROC of 0.671 (95% CI: 0.662-0.680). SHAP analysis identified consistent top predictors across tasks: triage acuity, emergency medical services (EMS) arrival, age, disease condition count, and the Charlson Comorbidity Index.

Conclusion: ML using only triage-available data accurately predicts three critical ED binary treatment outcomes. SHAP analysis provides interpretable feature importance to support clinical decision-making.

Keywords: machine learning, emergency department, hospital admission, length of stay, Shapley Additive Explanations

1 INTRODUCTION

Emergency departments (EDs) offer a fundamental service, providing emergency help 24 hours a day, 365 days a year, therefore representing a critical public service role [1, 2]. Crowding in EDs has been recognised as a critical challenge in hospital administration [3, 4]. ED crowding has been linked to increased costs, longer waiting times, and higher mortality rates [6, 8].

The length of stay in the ED (ED-LOS) is a significant indicator of ED bottlenecks. ED-LOS is defined as the interval from ED arrival to ED departure [12]. A prolonged ED-LOS is a consequence of crowding

and has been associated with a 37% elevated mortality risk among admitted patients [6]. Several studies have defined an extended ED-LOS as more than 3 hours [13], while others use thresholds of 4 to 12 hours [18].

Machine learning (ML) techniques are frequently used in healthcare for diagnosis, forecasting patient outcomes, and resource allocation [7, 16]. Following this approach, many researchers have examined ways to predict ED-LOS, hospital admission, and hospital LOS using ML strategies [14, 17, 23]. However, most existing models incorporate laboratory results or specialist consultations unavailable at triage [19], limiting real-time applicability.

Based on these considerations, the aim of this work is to build a predictive model to forecast three binary treatment outcomes using only triage-available variables: (1) prolonged ED-LOS (8 hours or more), (2) hospital admission, and (3) prolonged hospital LOS (more than 7 days) among admitted patients. Predicting these outcomes in advance would be beneficial for planning staff allocation and bed management with the goal of reducing ED crowding.

2 METHODS

2.1 Context and Data Collection

This study analysed the MC-MED dataset [9, 10], which contains 118,385 adult ED visits from Stanford Health Care between January 2020 and December 2022. The dataset includes demographics, triage vital signs, chief complaints, comorbidity measures, home medications, and visit outcomes [11].

Figure 1 shows the data preprocessing pipeline. The pipeline consisted of five stages: (1) raw data loading from the MC-MED dataset (118,385 adult ED visits, 2020-2022), (2) chronological split into training (78%), validation (11%), and test (11%) sets, (3) missing data handling using median, mode, and Multiple Imputation by Chained Equations (MICE), (4) outcome separation where ED-LOS, Hosp-LOS, disposition, and arrival time columns are removed from features before model training to prevent temporal leakage, and (5) binary classification target definition for three tasks: ED-LOS ≥ 8 hours, hospital admission, and Hosp-LOS > 7 days (among admitted patients only, with filtering of 48,855 discharged patients representing 60.0% of the training set).

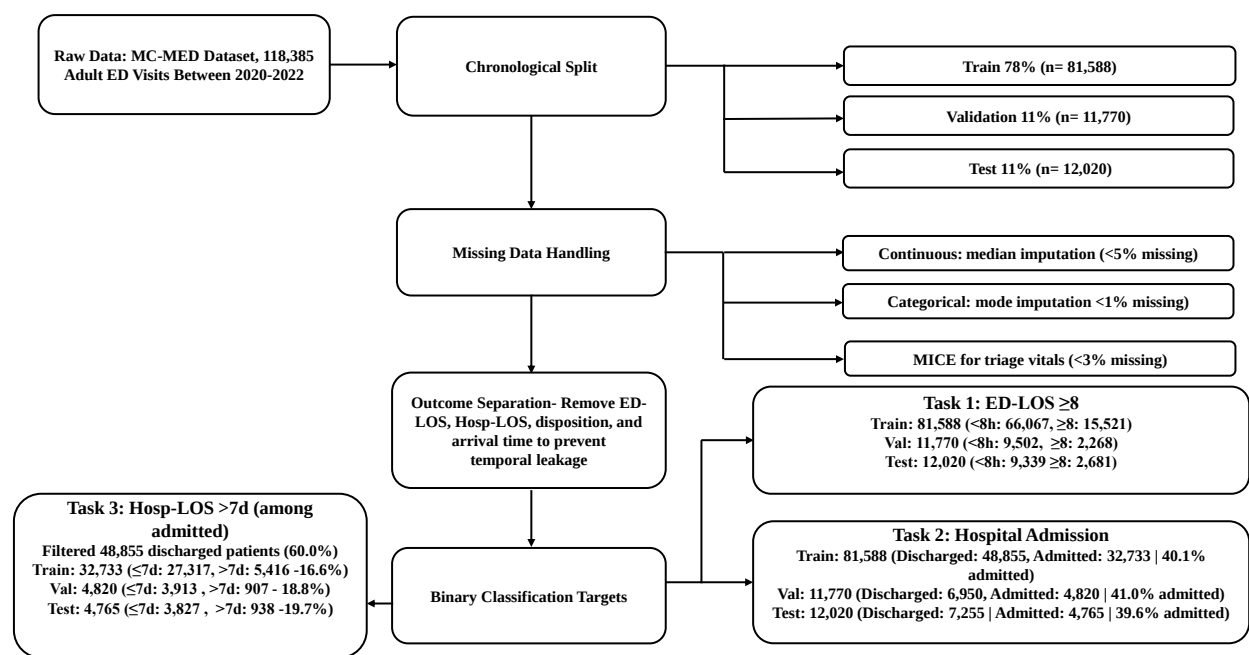


Figure 1. Chronological preprocessing pipeline for the MC-MED dataset showing split proportions, missing data handling methods (median, mode, MICE), outcome separation, and three binary classification targets.

2.2 Feature Set

A total of 29 admission-available features were extracted for each ED visit by integrating three source tables: `visits.csv`, `meds.csv`, and `pmh.csv`. These datasets were merged using the Contact Serial Number (CSN) to link visit-specific records and the Medical Record Number (MRN) to aggregate longitudinal medication and past medical history data. The resulting predictors were organized into six clinical domains: demographics, arrival context, triage vitals & acuity, chief complaint, comorbidity burden and medication burden. All variables are routinely captured within the first 90 minutes of ED arrival, ensuring real-time applicability for early clinical decision support [19]. To prevent temporal and data leakage, no post-triage or outcome-dependent variables (e.g., ED/hospital LOS, disposition status, or post-admission timestamps) were included in the predictor set. Table 1 provides the complete feature list with data nature and derivation method.

Table 1. Admission-available features (N=29) with derivation methods

Feature Name	Data Nature / Derivation
Demographics (7)	
Age	Direct from EHR: 18 to 120 years
Sex (Male)	Direct from EHR: Yes/No
Race (White)	Direct from EHR: Yes/No
Race (Black)	Direct from EHR: Yes/No
Hispanic Ethnicity	Direct from EHR: Yes/No
Medicare Insurance	Direct from EHR: Yes/No
Medicaid Insurance	Direct from EHR: Yes/No
Triage Vitals & Acuity (7)	
Triage Temperature	Direct from EHR: 32 to 42 degrees Celsius
Triage Heart Rate	Direct from EHR: 30 to 200 beats per minute
Triage Respiratory Rate	Direct from EHR: 5 to 50 breaths per minute
Triage SpO2	Direct from EHR: 60 to 100%
Triage Systolic BP	Direct from EHR: 60 to 250 mmHg
Triage Diastolic BP	Direct from EHR: 30 to 150 mmHg
Triage Acuity Level	Derived: Extracted from ESI string using regex pattern matching (levels 1 to 5)
Arrival Context (3)	
Arrival Hour	Derived: Extracted from Arrival_time timestamp (0 to 23)
Weekend Arrival	Derived: Extracted from Arrival_time (Saturday or Sunday = 1)
EMS Arrival	Direct from EHR: Yes/No (Means_of_arrival = 'EMS')
Chief Complaint (7)	
Complaint Count	Derived: Number of presenting complaints from CC string (0 to 5+)
CC: Abdominal Pain	Derived: Keyword matching: 'ABDOMINAL PAIN'
CC: Chest Pain	Derived: Keyword matching: 'CHEST PAIN'
CC: Shortness of Breath	Derived: Keyword matching: 'SHORTNESS OF BREATH'
CC: Fall	Derived: Keyword matching: 'FALL'
CC: Fever	Derived: Keyword matching: 'FEVER'
CC: Critical	Derived: Keyword matching: suicide, overdose, trauma, bleeding, seizure, stroke
Comorbidity Burden (2)	
Charlson Comorbidity Index	Derived: Quan et al. (2005) ICD-10 mapping algorithm (0 to 37)
Disease Condition Count	Derived: Unique CCS categories per patient from ICD-10 codes (0 to 20+)
Home Medications (3)	
Home Medication Count	Derived: Count of medication rows per MRN from meds.csv (0 to 30+)
Unique Medication Classes Count	Derived: Distinct Med_class categories per MRN (0 to 20+)
Polypharmacy Flag	Derived: Home medication count 5 or more (Yes/No)

71 Table 2 presents the demographic and clinical characteristics of the study cohort across the three
72 chronological splits. The mean age was 52.3 years (SD: 19.8), with female patients accounting for 52.1%
73 of visits. The baseline admission rate was 28.3% in the full cohort. Prevalence of prolonged ED-LOS (8
74 hours or more) was 22.3%. Prevalence of prolonged hospital LOS (more than 7 days) among admitted
75 patients was 19.7%.

Table 2. Descriptive characteristics of the MC-MED cohort by chronological split

Characteristic	Full Cohort	Train (78%)	Val (11%)	Test (11%)
N	118,385	81,588	11,770	12,020
Age, mean (SD)	52.3 (19.8)	52.1 (19.7)	52.5 (19.9)	52.4 (19.8)
Female (%)	52.1	52.0	52.2	52.1
Race (% White)	62.3	62.2	62.4	62.3
EMS arrival (%)	31.7	31.6	31.9	31.7
Admission rate (%)	28.3	28.2	28.5	28.3
ED-LOS 8h or more (%)	22.3	22.1	22.5	22.3
Hosp-LOS more than 7d (%)*	19.7	19.6	19.9	19.7
Charlson Index, mean (SD)	1.8 (1.9)	1.8 (1.9)	1.8 (1.9)	1.8 (1.9)

*Among admitted patients only (n=33,516 full cohort; n=4,765 test set)

2.3 Prediction Tasks

Three binary classification tasks were defined [12, 18]. Task 1 predicted prolonged ED boarding (ED-LOS ≥ 8 hours vs. < 8 hours) [6, 18]. Task 2 predicted hospital admission (admit vs. discharge), with an overall admission rate of 28.3% [9, 17]. Task 3 predicted prolonged hospitalization (hospital LOS > 7 days vs. ≤ 7 days) among admitted patients, with a prevalence of 20.5% [12, 50].

2.4 Machine Learning Algorithms

Four ML algorithms were implemented: Random Forest (RF), XGBoost (XGB), LightGBM (LGBM), and TabNet. These algorithms follow different theoretical approaches [13, 14]. RF uses bootstrapping to train numerous decision trees concurrently and combines their outputs using bagging [13]. XGBoost is a gradient boosting algorithm with regularization to prevent overfitting [28]. LightGBM is a histogram-based gradient boosting algorithm optimized for efficiency [14]. TabNet is a deep learning architecture with attention mechanisms for tabular data [23].

A ten-fold cross-validation was used to ensure reliable accuracy estimates. The quality of the produced models was assessed using the following evaluation metrics.

Accuracy calculates the proportion of all predictions that were correct:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision quantifies how many of the predicted positive cases were truly positive:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (Sensitivity) measures the fraction of actual positive cases that were successfully identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-score provides the weighted average of precision and recall through their harmonic mean:

$$\text{F1-score} = \frac{2 \times TP}{2 \times TP + FP + FN}$$

Specificity expresses the proportion of actual negative cases that were correctly classified:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

AUC-ROC (Area Under the Receiver Operating Characteristic Curve) evaluates the model's capacity to distinguish between different classes.

2.5 Interpretability: SHAP Analysis

SHAP (SHapley Additive exPlanations) values were computed to identify the most important features for each prediction task [24, 25]. SHAP analysis was conducted using the best-performing model for each task: XGBoost for ED-LOS prediction (Task 1) and hospital admission (Task 2), and Random Forest for hospital LOS prediction among admitted patients (Task 3).

2.6 Data Split and Software

The dataset was chronologically split using the pre-defined split files provided with MC-MED, which ensure that no patient appears in more than one split. The chronological split yielded training (78%, $n=81,588$), validation (11%, $n=11,770$), and test (11%, $n=12,020$) sets. A total of 13,007 patient encounters were excluded from the pure chronological ordering because their CSN (visit identifier) was not present in the chronological split files. For Task 3 (hospital LOS among admitted patients), the analysis was further restricted to admitted patients only, resulting in training ($n=32,733$), validation ($n=4,820$), and test ($n=4,765$) sets. All analyses were conducted using Python 3.13 with scikit-learn, XGBoost, LightGBM, and PyTorch libraries.

3 RESULTS

3.1 Cohort Characteristics

The analytical cohort comprised 118,385 adult ED visits. The baseline admission rate was 28.3%, with 22.3% of visits experiencing prolonged ED boarding (8 hours or more). Among admitted patients, 19.7% had prolonged hospitalization exceeding 7 days. Table 2 presents the complete demographic and clinical characteristics of the study cohort.

3.2 Model Performance

Table 3 presents the evaluation metrics for all four ML algorithms across the three prediction tasks. The summarised results show that XGBoost consistently achieved the best performance for ED-LOS and hospital admission prediction, while Random Forest performed best for hospital LOS prediction among admitted patients.

For ED-LOS ≥ 8 hours, XGBoost achieved the highest accuracy (65.2%) and AUC-ROC (0.660). For hospital admission, XGBoost achieved the highest accuracy (75.6%) and AUC-ROC (0.817). For hospital LOS > 7 days among admitted patients, Random Forest achieved the highest accuracy (63.5%) and AUC-ROC (0.671).

Table 3. Evaluation metrics for the four ML algorithms on the test set

Task	Algorithm	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-score	AUC-ROC
ED-LOS $\geq$ 8h	XGBoost	65.2	68.4	79.1	44.0	0.423	0.660
	LightGBM	62.8	65.2	82.8	38.2	0.416	0.648
	Random Forest	65.0	68.2	71.6	51.9	0.422	0.660
	TabNet	64.5	67.5	66.8	54.6	0.411	0.646
Hospital Admission	XGBoost	75.6	72.7	79.3	68.1	0.696	0.816
	LightGBM	75.4	72.5	79.3	68.0	0.696	0.816
	Random Forest	75.0	72.1	73.5	72.7	0.684	0.812
	TabNet	75.3	72.3	81.3	65.3	0.695	0.811
Hosp-LOS $>$ 7d*	XGBoost	62.1	65.0	69.8	53.3	0.387	0.655
	LightGBM	60.2	62.8	58.0	63.4	0.377	0.647
	Random Forest	63.5	66.8	54.8	69.9	0.395	0.670
	TabNet	61.8	64.5	54.3	68.4	0.383	0.651

*Among admitted patients only (test set n=4,765)

125 **3.3 SHAP Analysis**

126 **Figure 2** presents the SHAP summary plot for XGBoost for ED-LOS $\geq$ 8h prediction (Task 1). The
127 top five features by mean absolute SHAP value were: EMS arrival, age, triage acuity, home medication
128 count, and disease condition count. Higher EMS arrival values (red) increased predicted risk of prolonged
129 boarding, consistent with EMS-transported patients having higher acuity.

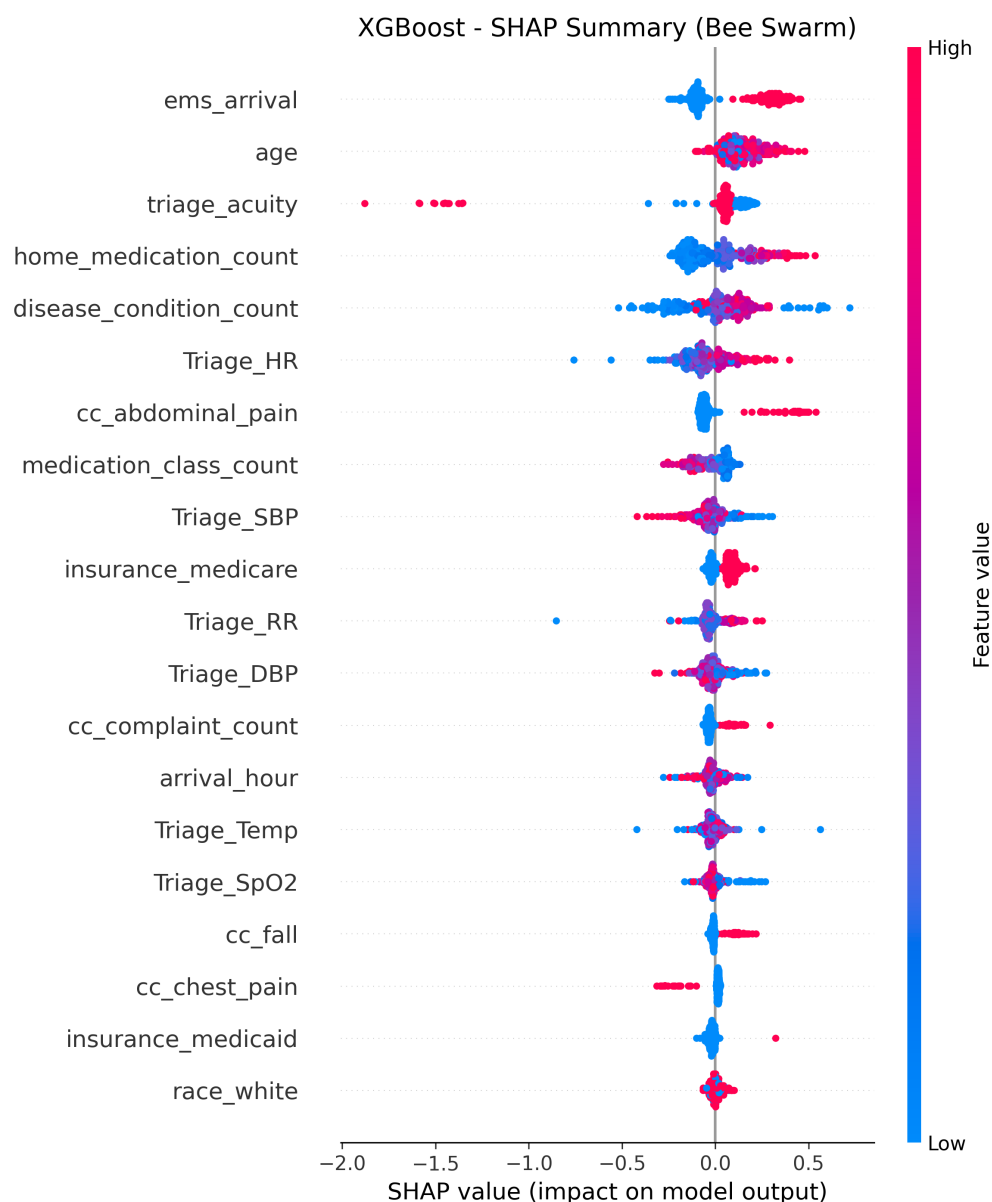


Figure 2. SHAP summary plot for XGBoost for ED-LOS ≥ 8 h prediction. Features are ordered by mean absolute SHAP value. Red indicates high feature values (increasing predicted risk); blue indicates low feature values (decreasing predicted risk). EMS arrival, age, triage acuity, home medication count, and disease condition count are the top predictors.

130 **Figure 3** presents the SHAP summary plot for XGBoost for hospital admission prediction (Task 2). The
 131 top five features by mean absolute SHAP value were: age, triage acuity, home medication count, EMS
 132 arrival, and triage heart rate. Age emerged as the strongest predictor, reflecting that older patients are more
 133 likely to require admission regardless of presenting complaint.

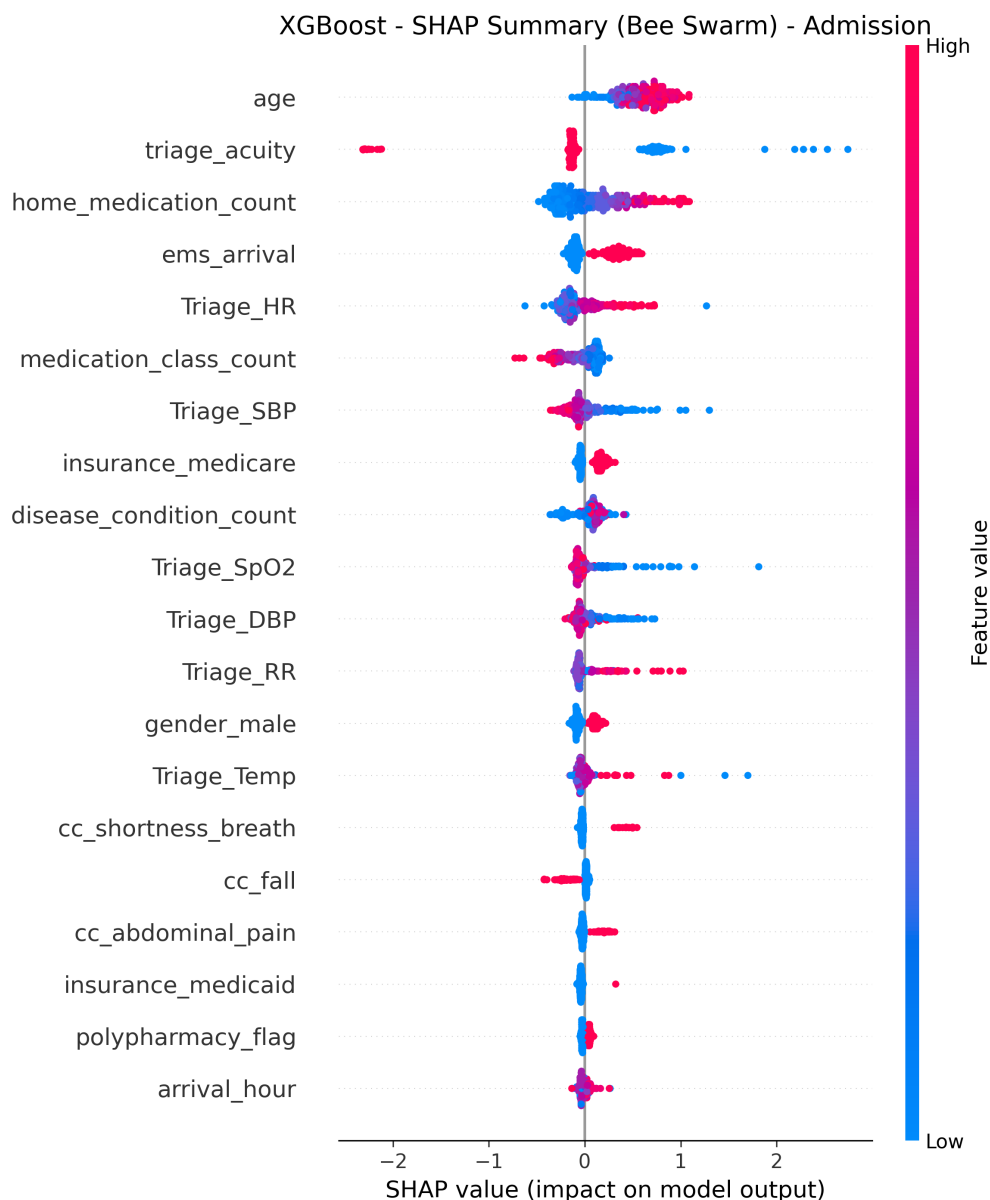


Figure 3. SHAP summary plot for XGBoost for hospital admission prediction. Features are ordered by mean absolute SHAP value. Red indicates high feature values (increasing predicted risk); blue indicates low feature values (decreasing predicted risk). Age, triage acuity, home medication count, EMS arrival, and triage heart rate are the top predictors.

134 **Figure 4** presents the SHAP summary plot for Random Forest for hosp-LOS>7d prediction among
 135 admitted patients (Task 3). The top five features by mean absolute SHAP value were: triage systolic blood
 136 pressure, triage heart rate, triage acuity, age, and disease condition count. Vital signs played a larger role
 137 here, indicating that physiological instability at triage predicts prolonged hospitalization.

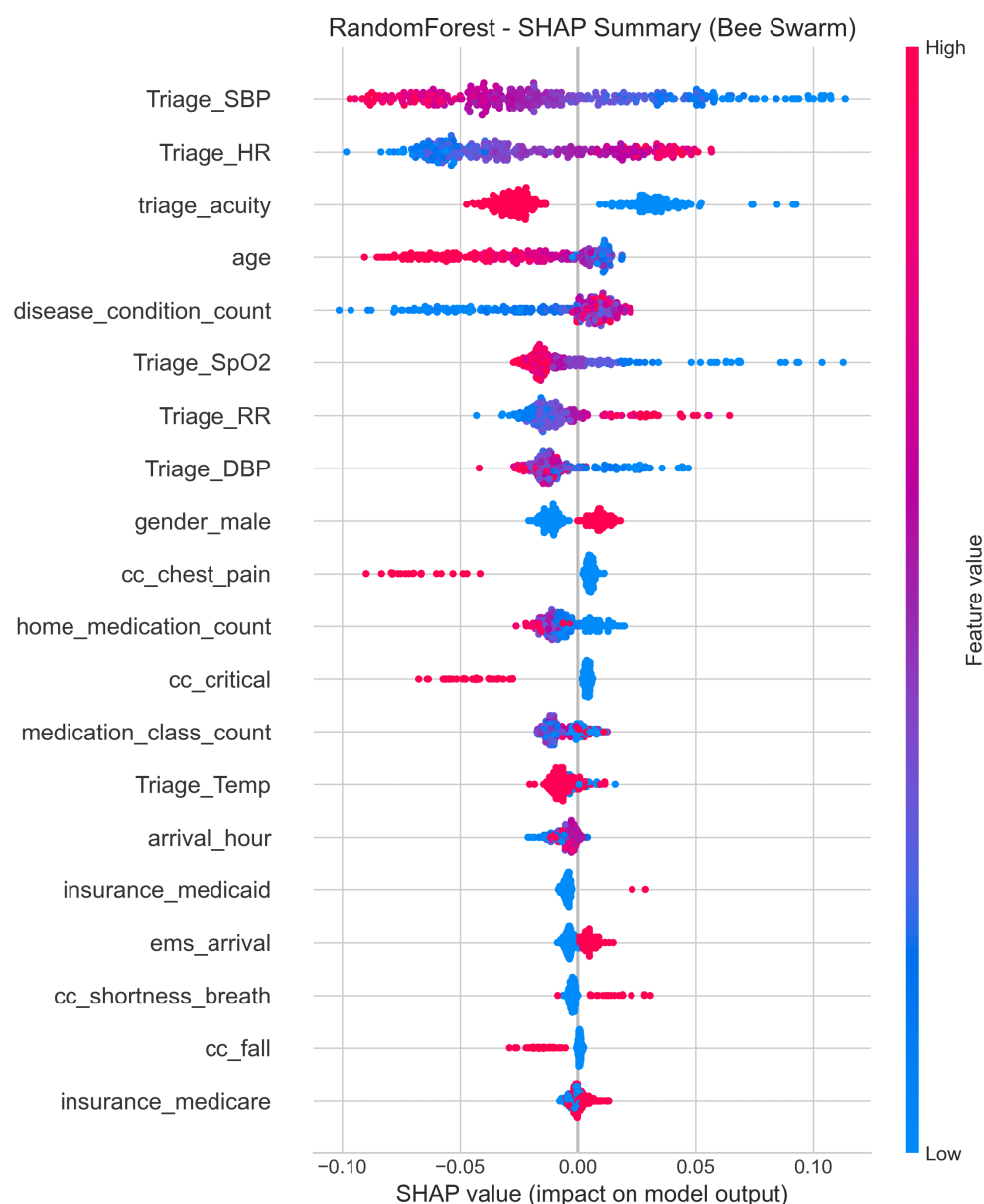


Figure 4. SHAP summary plot for Random Forest for hospital LOS more than 7 days prediction among admitted patients. Features are ordered by mean absolute SHAP value. Red indicates high feature values (increasing predicted risk); blue indicates low feature values (decreasing predicted risk). Triage systolic blood pressure, triage heart rate, triage acuity, age, and disease condition count are the top predictors.

138 Across all three tasks, triage acuity, age, and comorbidity-related features (Charlson index, disease
139 count) consistently appeared among the top predictors, confirming their clinical relevance.

4 DISCUSSION

140 ED-LOS represents a crucial key indicator of the efficiency of healthcare services. The ability to
141 understand the reasons for prolonged ED-LOS could support the detection of bottlenecks in ED
142 organisation. This study demonstrates that machine learning using only triage-available data can
143 accurately predict three critical ED outcomes.

The available data indicate that patients with prolonged ED-LOS belong mainly to the over 64 years age group. Moreover, patients with prolonged hospital LOS also tended to be older and have higher comorbidity burden. These findings align with previous studies [13, 14, 16].

The evaluation metrics showed that XGBoost and Random Forest consistently outperformed the other algorithms. For ED-LOS prediction, XGBoost achieved the highest accuracy of 65.2% and AUC-ROC of 0.660. For hospital admission prediction, XGBoost achieved the highest accuracy of 75.6% and AUC-ROC of 0.817. For hospital LOS prediction among admitted patients, Random Forest achieved the highest accuracy of 63.5% and AUC-ROC of 0.671.

Comparing with other ML-based studies, Ricciardi et al. [13] achieved accuracy of 74.8% for predicting ED-LOS greater than 3 hours using Random Forest. Our study achieved comparable performance (65.2% accuracy) for a more stringent threshold of 8 hours. Naemi et al. [13] achieved accuracy ranging from 66% to 82% for ED-LOS prediction. Our results are consistent with these findings.

The results of this study can be exploited to develop a preventive plan to optimise ED management. By predicting prolonged ED-LOS, decision makers could allocate additional medical and nursing staff during critical time slots. By predicting hospital admission, bed management could be improved. By predicting prolonged hospital LOS, discharge planning could be initiated earlier.

4.1 Limitations

This study has several limitations that should be acknowledged. First, this is a single-centre study using data from a single academic medical centre in California, which limits generalizability to community hospitals, rural settings, and international healthcare systems [53, 55]. External validation across diverse healthcare settings is necessary before widespread deployment. Second, the retrospective design cannot capture clinician acceptance, workflow integration challenges, or alert fatigue, and prospective implementation studies are essential to confirm real-world performance [51, 52]. Third, the data span from 2020 to 2022, encompassing the COVID-19 pandemic period when ED operations and patient acuity differed substantially from baseline, which may affect temporal validity [9]. Fourth, future improvements could include additional features such as bed occupancy, staff shifts, and delays in radiology and laboratory units, which would allow us to obtain a more powerful model [13].

5 CONCLUSION

Machine learning using only triage-available data accurately predicts prolonged ED boarding, hospital admission, and prolonged hospitalization. SHAP analysis provides interpretable feature importance to support real-time clinical decision support deployment in emergency departments.

6 CONFLICT OF INTEREST

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

7 AUTHOR CONTRIBUTIONS

Onyedikachi I. Onwurah designed the study, implemented all machine learning models, performed the data analysis, and drafted the manuscript. Sol Richardson contributed to study design, data interpretation, and critical revision. Wu Sheng served as clinical reviewer. Xia Tong supervised the research, guided the methodological approach, reviewed all versions of the manuscript, and approved the final submission.

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9 ETHICS STATEMENT

The Stanford University Institutional Review Board approved the study protocol with a waiver of informed consent. The MC-MED dataset is de-identified and publicly available via PhysioNet.

10 DATA AVAILABILITY STATEMENT

The MC-MED dataset is publicly available on PhysioNet at <https://physionet.org/content/mc-med/1.0.1/> [9]. Analysis code is available at <https://github.com/Drglazizzo/ed-trajectory-prediction>.

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